



PENRITH HIGH SCHOOL

**2013
HSC TRIAL EXAMINATION**

Mathematics Extension 1

General Instructions:

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total marks–70

SECTION I

Pages 3–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

SECTION II

Pages 6–9

60 marks

- Attempt Questions 11–14
- Allow about 1 hours 45 minutes for this section

Student Name: _____

Teacher Name: _____

This paper MUST NOT be removed from the examination room

This page is for use by teachers ONLY

Student Name: _____

Teacher Name: _____

Exam Outcomes

1. Use the relationship between functions, inverse functions and derivatives (**Differential Calculus**)
2. Study of simple harmonic and projectile motion (**Motion**)
3. Manipulate polynomial functions (**Polynomials**)
4. Applies angle and chord properties of the circle (**Circle Geometry**)
5. Problem solving (**PS**)

Paper Grid

Outcome	Mark	Qn-Num/Out-of	Mark
Differential Calculus	/33	1 / 1 4 / 1 6 / 1 8 / 1 9 / 1 10 / 1	/6
		11-a / 1 11-e / 3	/4
		12-a / 1 12-b / 5 12-c / 2 12-e / 4	/12
		13-a / 2 13-d / 7	/9
		14-b / 2	/2
Motion	/13	14-a / 3 14-c / 10	/13
Polynomials	/5	2 / 1 7 / 1	/2
		13-b / 3	/3
Circle Geometry	/3	12-d / 3	/3
Problem Solving	/16	3 / 1 5 / 1	/2
		11-b / 2 11-c / 2 11-d / 2 11-f / 5	/11
		13-c / 3	/3
Exam Total	/70		%

Assessor: Daniel Antone

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10

- 1)! A point P moves in the xy -plane such that $P(\tan\theta, \cot\theta)$ is its parametric presentation with the parameter θ , where θ is any real number. The locus of P then is

- (A) Parabola
- (B) Circle
- (C) Hyperbola
- (D) Straight Line

- 2)! Let $P(x)$ be a polynomial of degree $n > 0$. Let $Q(x)$ be a polynomial of degree $m \leq n$ such that

$$P(x) = (x - a)^r Q(x) + R(x)$$

Then the degree of $R(x)$ is

- (A) $n + m + r$
- (B) $n - m - r$
- (C) $n + m - r$
- (D) $n - m + r$

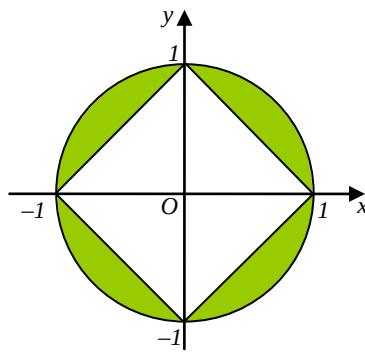
- 3)! The sum of this infinite geometric series $\sqrt{2} - 1 + \frac{1}{\sqrt{2}} - \frac{1}{2} + \frac{1}{2\sqrt{2}} - \dots$ is closest to

- (A) 0.5
- (B) 1
- (C) 1.5
- (D) 2

- 4)! Let $T(x)$ be a function defined by $T(x) = [f(x)g(x)]^{n+1}$, where $f(x)$ and $g(x)$ are two real functions. Then $\frac{dT}{dx}$ is

- (A) $\frac{dT}{dx} = (n+1)[f(x)g(x)]^n \frac{df}{dx} \cdot \frac{dg}{dx}$
 (B) $\frac{dT}{dx} = (n+1)[f(x)g(x)]^n \left[\frac{df}{dx} + \frac{dg}{dx} \right]$
 (C) $\frac{dT}{dx} = (n+1)[f(x)g(x)]^n \left[f \frac{df}{dx} + g \frac{dg}{dx} \right]$
 (D) $\frac{dT}{dx} = (n+1)[f(x)g(x)]^n \left[g \frac{df}{dx} + f \frac{dg}{dx} \right]$

- 5)! The only set of inequalities that represents the shaded regions between the circle and the square below is



- (A) $x^2 + y^2 \leq 1$ and $|x| + |y| \geq 1$
 (B) $x^2 + y^2 \leq 1$ and $|x| - |y| \geq 1$
 (C) $x^2 + y^2 \leq 1$ and $|x - y| \geq 1$
 (D) $x^2 + y^2 \leq 1$ and $|x + y| \leq 1$
- 6)! Consider the functions $f(x) = e^x$ and $g(x) = \ln x$. Let a be a real number such that $a > 1$. The only correct statement of the following is
- (A) $f'(a) \leq g'(a)$
 (B) $f'(a) \geq g'(a)$
 (C) $f'(a) < g'(a)$
 (D) $f'(a) > g'(a)$

7)! Let $f(x)$ be the cubic polynomial defined by $f(x) = (x - 1)^3 + x$. The point $(1, 1)$ is

- (A) a stationary point of $f(x)$
- (B) a turning point of $f(x)$
- (C) a horizontal point of inflexion of $f(x)$
- (D) a non-horizontal point of inflexion of $f(x)$

8)! The only correct statement about the function $f(\theta) = \frac{2 \sin(\theta + 45)}{5 \cos(45 - \theta)}$ is that

- (A) it is a constant function
- (B) it varies as θ varies
- (C) it has a maximum value 0.4
- (D) it has a minimum value 0.4

9)! The domain and range for the function $y = 2 \cos^{-1}(x)$ is

- (A) Domain: $-1 \leq x \leq 1$, Range: $0 \leq y \leq 2\pi$
- (B) Domain: $-1 \leq x \leq 1$, Range: $0 \leq y \leq \pi$
- (C) Domain: $0 \leq x \leq 1$, Range: $0 \leq y \leq 2\pi$
- (D) Domain: $0 \leq x \leq 1$, Range: $0 \leq y \leq \pi$

10)! Consider $f(x) = \ln(x) - \ln(-x)$. Then $f(x)$ is

- (A) An even function
- (B) An odd function
- (C) Undefined everywhere
- (D) A relation which is not a function

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations

Question 11 (15 marks) Start a NEW page

- (a)! Find all exact values of the angle $\tan^{-1}(-\sqrt{3})$ in radians **1**
- (b)! $A(-5, 6)$ and $B(1, 3)$ are two points. Find the coordinates of the point P which divides the interval AB externally in the ratio 5:2. **2**
- (c)! The perpendicular distance from the point (x_1, y_1) to the line $y = x + 3$ is $2\sqrt{2}$ and to x-axis is 3. Find the coordinates of the point (x_1, y_1) . **2**
- (d)! Find all possible solutions for the equation $(2x^2 - 1)^2 = \frac{(2x^2 - 1)^2}{(2 - 4x^2)}$. **2**
- (e)! Show that the first derivative of the function $2x\sqrt{\sin x}$ may be given as **3**

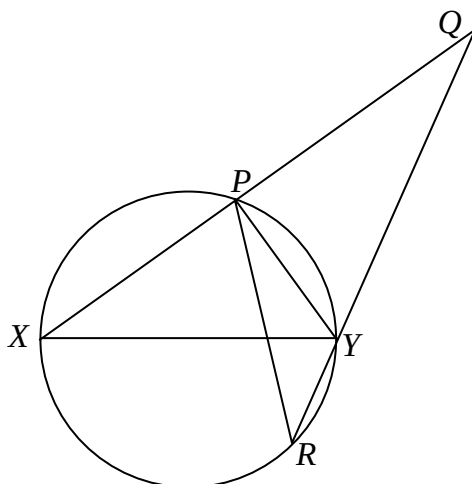
$$\frac{\cos x + 2 \sin x}{\sqrt{\sin x}}.$$
- (f)! Consider the following functions
- $$f(x) = x^2 + 4x - 12 \text{ and } g(x) = \frac{x+6}{2-x}$$
- (i) Solve the identity $f(x) \leq 0$ and indicate your solution on a number line. **2**
- (ii) Solve the identity $g(x) \leq 0$ and indicate your solution on a number line. **2**
- (iii) Find the simultaneous solution of the two inequalities $f(x) \leq 0$ and $g(x) \leq 0$. **1**

Proceed to next page for question (12)

Question 12 (15 marks) Start a NEW page

- (a)! A circular plate is being expanded by heating. When the radius just reaches a value of 20 cm, it (the radius) is increasing at the rate of 0.01 cm/s. Find the rate of increase in the area at this moment in terms of π . 1
- (b)! Consider the two exponential functions $y = e^{2x}$ and $y = e^x + 2$.
- (i) Draw a neat sketch showing the graphs of $y = e^{2x}$ and $y = e^x + 2$ on the same diagram, showing any asymptotes and axes intercepts. 2
- (ii) Show that the coordinates of the point of intersection of $y = e^{2x}$ and $y = e^x + 2$ is $(\ln 2, 4)$. 1
- (iii) Find the area bounded by the y -axis and the two curves $y = e^{2x}$ and $y = e^x + 2$. Give your final answer correct to 2 decimal places. 2
- (c)! The size of the acute angle between the tangents drawn to the curve $y = \ln x$ at the points where $x = 1$ and $x = x_1$ is $\frac{\pi}{6}$. Find the exact value of x_1 . 2

- (d)! 3



XY is a diameter in the circle above. Given that $\angle X = 35^\circ$ and $\angle Q = 25^\circ$, find the size of $\angle YPR$, giving reasons.

- (e)! Let $y = \sin^{-1}(1 - x^2)$.
- (i) By using the substitution $u = 1 - x^2$, or otherwise, show that $\frac{dy}{dx} = \frac{-2}{\sqrt{1 - x^2}}$. 1
- (ii) Hence show that $f'(x) = 0$ where $f(x) = 2 \cos^{-1}\left(\frac{x}{\sqrt{2}}\right) - \sin^{-1}(1 - x^2)$. 1
- (iii) Hence or otherwise, show that $2 \cos^{-1}\left(\frac{x}{\sqrt{2}}\right) - \sin^{-1}(1 - x^2) = \frac{\pi}{2}$. 2

Proceed to next page for question (13)

Question 13 (15 marks) Start a NEW page

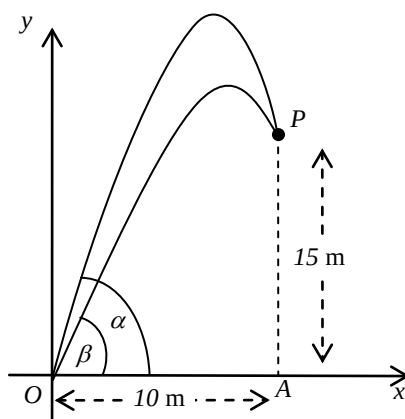
- (a)! Evaluate $\int_0^1 \sqrt{1-x^2} dx$ using the substitution $x = \sin \theta$ 2
- (b)! (i) State the conditions that the quadratic expression $ax^2 + bx + c$ is negative definite. 1
- (ii) Show that the expression $(k^2 + k)x^2 - (2k - 6)x + 2$, where $k \neq 0$, can never be negative definite. 1
- (iii) Find the range of values of k for which the expression is positive definite. 1
- (c)! Prove by mathematical induction that 3
- $$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1) \times n! = n \times (n+1)! \text{ for all integers } n = 1, 2, 3, \dots$$
- (d)! A machine which initially costs \$49 000 loses value at a rate proportional to the difference between its current value \$ M and its final scrap value \$1000. After 2 years the value of the machine is \$25 000.
- (i) Explain why $\frac{dM}{dt} = -k(M - 1000)$ for some constant $k > 0$, and verify that $M = 1000 + Ae^{-kt}$, A a constant, is a solution of this equation. 2
- (ii) Find the exact values of A and k . 3
- (iii) Find the value of the machine, and the time that has elapsed, when the machine is losing value at a rate equal to one quarter of the initial rate at which it loses value. 2

Proceed to next page for question (14)

Question 14 (15 marks) Start a NEW page

- (a)! A particle is moving with simple harmonic motion in a straight line. It has amplitude of 10 metres and a period of 10 seconds.
- (i) Prove that it would take the particle $\frac{5}{\pi} \cos^{-1} \frac{3}{5}$ sec to travel from one of the extremities of its path to a point 4 metres away? **2**
- (ii) At what speed, correct to whole m/s, would the particle reach this position? **1**
- (b)! It is known that $\ln x + \sin x = 0$ has a root close to $x = 0.5$. Use one application of Newton's Method to obtain a better approximation of the root to 4 decimal places. **2**
- (c)! A projectile with initial velocity $U \text{ ms}^{-1}$ at an angle of projection α , and acceleration downwards due to gravity, g , has been fired from the origin.
- (i) At a time $t \geq 0$ seconds the projectile is at the point (x, y) , prove that **3**
- $$x = Ut \cos \alpha \quad \text{and} \quad y = Ut \sin \alpha - \frac{1}{2}gt^2$$
- (ii) Show that the equation of the path of a projectile is given by **2**
- $$y = x \tan \alpha - \frac{gx^2}{2U^2} \sec^2 \alpha$$

Nicholas throws a small pebble from a fixed point O on level ground, with a velocity $U = 7\sqrt{10} \text{ ms}^{-1}$ at an angle α , with the horizontal. Shortly afterwards, he throws another small pebble from the same point at the same speed but at a different angle to the horizontal β , where $\beta < \alpha$, as shown. The pebbles collided at a point $P(10, 15)$. Consider the acceleration downwards due to gravity is $g = 9.8 \text{ ms}^{-2}$.



- (iii) Show that the two possible initial angles of projection are **3**
- $$\alpha = \tan^{-1} 8 \text{ and } \beta = \tan^{-1} 2$$
- (iv) Show that the time elapsed between when the pebbles were thrown **2**
- $$\text{was } \frac{\sqrt{650} - \sqrt{50}}{7} \text{ seconds.}$$

End of paper

[End Of Qns]

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a \neq 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, x > 0$

HSC - Trial Exam (1)
2013

Mult. Choice Answers

Section (I)

1) C

6) D

2) B

7) D

3) B

8) A

4) D

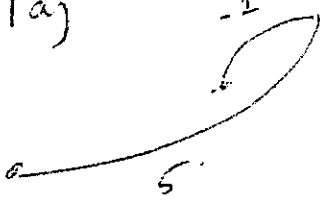
9) A

5) A

10) C

(2)

11a)



$$\frac{5 \times 1 + -2 \times -5}{3}, \frac{5 \times 3 + -2 \times 6}{3}$$

$$= \frac{15}{3}, \frac{3}{3}$$

$$= (5, 1)$$

$$b) x - y + 3 = 0$$

perp distance

$$\left| \frac{-1x_1 - 1y_1 + 3}{\sqrt{2}} \right| = 2\sqrt{2}$$

$$x - y + 3 = \pm 4$$

$$y = 3 \rightarrow \text{from distance to } x \text{ axis is } 3$$

$$x - y = \pm 1$$

$$\text{when } x = 4 \quad y = \pm 3$$

$$\text{when } y = -3 \quad x = -10$$

$$\text{or } x = -2$$

$$y = 3 \quad x = -4$$

$$\text{or } x = 4$$

$$(\pm 4, 3), (-2, -3), (-10, -3)$$

$$(c) (2x^2 - 1)^2 = \frac{(2x^2 - 1)^2}{(2 - 4x^2)}$$

$$(2x^2 - 1)^2 (2 - 4x^2) - (2x^2 - 1)^2 = 0$$

$$(2x^2 - 1)^2 [2 - 4x^2 - 1] = 0$$

$$(2x^2 - 1)^2 [1 - 4x^2] = 0$$

$$(2x^2 - 1) = 0 \quad \text{or} \quad (1 - 4x^2) = 0$$

$$x^2 = \frac{1}{2}$$

$$x = \pm \frac{1}{\sqrt{2}}$$

$$(1 - 2x)(1 + 2x) = 0$$

$$x = \pm \frac{1}{2}$$

$$d) 2x (\sin x)^{\frac{1}{2}} \quad \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

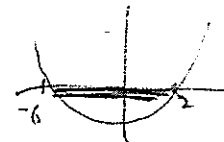
$$(\sin x)^{\frac{1}{2}} \cdot 2 + 2x \cdot \frac{1}{2} (\sin x)^{-\frac{1}{2}} \cdot \cos x$$

$$= 2\sqrt{\sin x} + \frac{x \cos x}{\sqrt{\sin x}}$$

$$= \frac{2 \sin x + x \cos x}{\sqrt{\sin x}}$$

$$e) (i) x^2 + 4x - 12 \leq 0$$

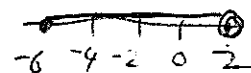
$$(x + 6)(x - 2) \leq 0$$



$$\therefore -6 \leq x \leq 2$$

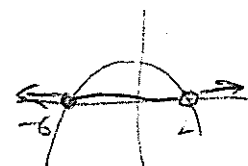
undefined at $x = 2$

$$\therefore -6 \leq x < 2$$



$$(ii) \frac{x+6}{2-x} \leq 0$$

$$(x+6)(2-x) \leq 0$$



$$\therefore x \leq -6 \text{ or } x > 2$$

$$(iii) \text{ Simultaneous } x = -6$$

$$(iv) f(x) = \frac{x^3}{x}$$

$$g(x) = -\frac{x^3}{x}$$

12)

$$a) A = \pi r^2$$

$$\frac{dA}{dr} = 2\pi r$$

$$\frac{dr}{dt} = 0.01$$

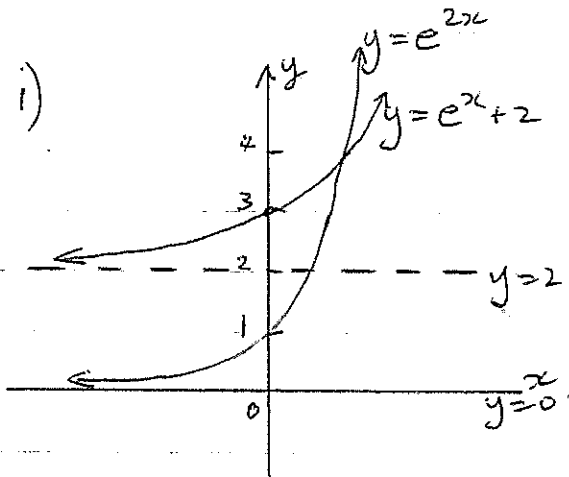
$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$r = 20$$

$$\frac{dA}{dr} = 2\pi \times 20 \times 0.01$$

$$= 0.4\pi \text{ cm}^2/\text{sec}$$

b) i)



$$\begin{aligned} \text{ii)} \quad e^{2x} &= e^x + 2 \\ e^{2x} - e^x - 2 &= 0 \\ (e^x)^2 - e^x - 2 &= 0 \\ (e^x - 2)(e^x + 1) &= 0 \end{aligned}$$

$$\therefore e^x = 2 \quad \text{or} \quad e^x = -1$$

$$x = \ln 2 \quad \text{No. sol}^n$$

$$y = e^{2 \ln 2} = e^{\ln 2^2}$$

$$= e^{\ln 4}$$

$$= 4 \quad \therefore (\ln 2, 4)$$

iii)

$$A = \int_0^{\ln 2} [e^x + 2 - e^{2x}] dx$$

$$= \left[e^x + 2x - \frac{e^{2x}}{2} \right]_0^{\ln 2}$$

$$= \left[e^{\ln 2} + 2 \ln 2 - \frac{e^{2 \ln 2}}{2} \right] - \left[e^0 + 0 - \frac{e^0}{2} \right]$$

$$= \left(2 + 2 \ln 2 - \frac{4}{2} \right) - \left(1 - \frac{1}{2} \right)$$

$$= 2 \ln 2 - \frac{1}{2}$$

$$\doteq 0.89 \text{ (2 sig. fig)}$$

c)

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$y = \ln x \quad \therefore \frac{dy}{dx} = \frac{1}{x}$$

$$\text{for } x = 1, \quad m_1 = 1$$

$$\text{for } x = x_1, \quad m_2 = \frac{1}{x_1}$$

$$\therefore \tan\left(\frac{\pi}{6}\right) = \left| \frac{1 - \frac{1}{x_1}}{1 + 1 \cdot \frac{1}{x_1}} \right|$$

$$\frac{1}{\sqrt{3}} = \left| \frac{\frac{x_1 - 1}{x_1}}{\frac{x_1 + 1}{x_1}} \right|$$

$$\frac{1}{\sqrt{3}} = \left| \frac{x_1 - 1}{x_1 + 1} \right|$$

$$\therefore \frac{x_1 - 1}{x_1 + 1} = \frac{1}{\sqrt{3}} \quad \text{or} \quad \frac{x_1 - 1}{x_1 + 1} = -\frac{1}{\sqrt{3}}$$

$$\sqrt{3}x_1 - \sqrt{3} = x_1 + 1$$

$$\sqrt{3}x_1 - \sqrt{3} = -x_1 - 1$$

$$\sqrt{3}x_1 - x_1 = \sqrt{3} + 1$$

$$\sqrt{3}x_1 + x_1 = \sqrt{3} - 1$$

$$x_1 = \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

$$x_1 = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$= 2 + \sqrt{3}$$

$$= 2 - \sqrt{3}$$

(4)

12)

d) i)

$$y = \sin^{-1}(1-x^2) \quad \text{Let } u = 1-x^2$$

$$y = \sin^{-1}(u) \quad \frac{du}{dx} = -2x$$

$$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$$

$$= \frac{1}{\sqrt{1-(1-x^2)^2}}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{1}{\sqrt{1-(1-2x^2+x^4)}} \times -2x$$

$$= \frac{-2x}{\sqrt{1-1+2x^2-x^4}}$$

$$= \frac{-2x}{\sqrt{x^2(2-x^2)}}$$

$$= \frac{-2x}{x\sqrt{2-x^2}}$$

$$= \frac{-2}{\sqrt{2-x^2}}$$

ii)

$$f(x) = 2 \cos^{-1}\left(\frac{x}{\sqrt{2}}\right) - \sin^{-1}(1-x^2)$$

$$f'(x) = 2 \cdot \frac{-1}{\sqrt{(\sqrt{2})^2 - x^2}} \cdot \frac{-2}{\sqrt{2-x^2}}$$

$$= \frac{-2 \cdot -2}{\sqrt{2-x^2} \sqrt{2-x^2}} + \frac{2}{\sqrt{2-x^2}}$$

$$= 0$$

iii) If $f'(x) = 0$ then

$y = f(x)$ is a horizontal line.

$$\text{i.e. } y = c$$

sub any x value to find c .

$$f(0) = 2 \cos^{-1}\left(\frac{0}{\sqrt{2}}\right) - \sin^{-1}(1-0^2)$$

$$= 2 \cos^{-1}(0) - \sin^{-1}(1)$$

$$= 2 \times \frac{\pi}{2} - \frac{\pi}{2}$$

$$= \frac{\pi}{2}$$

$$\therefore f(x) = \frac{\pi}{2}$$

Q13 Ext 1 2013

5

a) $\tan^{-1}(-\sqrt{3})$  $\frac{S}{T} = \frac{A}{C}$ 0 to 2π .

$$= \pi - \frac{\pi}{3} \text{ or } 2\pi - \frac{\pi}{3} = \boxed{\frac{2\pi}{3}} \text{ or } \boxed{\frac{5\pi}{3}}$$

b) $\int_0^1 \sqrt{1-x^2} dx$ $x = \sin \theta$

$$= \int \sqrt{1-\sin^2 \theta} \cos \theta d\theta$$

$$\frac{dx}{d\theta} = \cos \theta$$

$$dx = \cos \theta d\theta$$

$$= \int \sqrt{\cos^2 \theta} \cos \theta d\theta$$

$$= \int \cos \theta \cdot \cos \theta d\theta = \int \cos^2 \theta d\theta$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 + \cos 2\theta d\theta$$

As $x=0, \theta=0$
~~sin~~ $x=1, \theta=\frac{\pi}{2}$

$$= \frac{1}{2} \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{2}} = \frac{1}{2} \left(\left(\frac{\pi}{2} + \frac{\sin \pi}{2} \right) - 0 \right)$$

$$= \frac{1}{2} \times \frac{\pi}{2} = \boxed{\frac{\pi}{4}}$$

c) i) ax^2+bx+c neg def

$a < 0, \Delta = b^2 - 4ac < 0$



(ii) $a = k^2 + k$ $b = 6 - 2k$ $c = 2$

$$\Delta = b^2 - 4ac = (6-2k)^2 - 4 \times (k^2+k) \times 2$$

$$= 36 - 24k + 4k^2 - 8k^2 - 8k$$

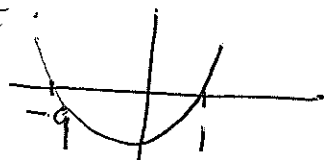
$$= -4k^2 - 32k + 36$$

$$= -4(k^2 + 8k - 9)$$

$$= -4(k+9)(k-1) \text{ Solve } < 0$$

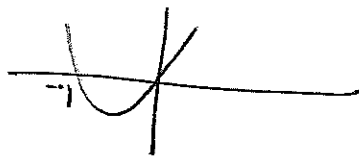
$$\therefore \text{Solve } (k+9)(k-1) > 0$$

(ii) cont.



$$k > 1 \text{ or } k < -9 \quad \textcircled{A}$$

Also $a = k^2 + k < 0$
 $k(k+1) < 0$



$$-1 < k < 0 \quad \textcircled{B}$$

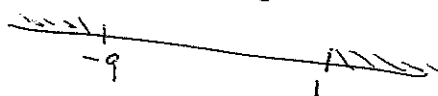
Both $\textcircled{A}$ and $\textcircled{B}$ must hold - no simultaneous solution. $\therefore$ cannot be negative definite.

(iii) +ve def $a > 0, \Delta < 0$

$$k(k+1) > 0 \quad \begin{cases} k > 0 \text{ or } k < -1 \end{cases}$$

$$\Delta < 0 \quad \begin{cases} k > 1 \text{ or } k < -9 \text{ (from above)} \end{cases}$$

Solve simultaneously



$$\therefore k > 1 \text{ or } k < -9$$

d) $2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1) \times n! = n(n+1)!$ $\textcircled{1}$

Step 1 Show Statement true for $n=1$

$$\text{LHS of } \textcircled{1} = (1^2 + 1) \times 1! = 2$$

$$\text{RHS of } \textcircled{1} = 1(1+1)! = 2$$

$$\text{LHS} = \text{RHS}$$

$\therefore$ Statement true for $n=1$

Step 2 Assume Statement true for $n=k$, i.e.

$$\text{assume } 2 \times 1! + 5 \times 2! + \dots + (k^2 + 1) \times k! = k(k+1)!$$

Prove statement is true for $n=k+1$, i.e. prove

$$2 \times 1! + 5 \times 2! + \dots + (k^2 + 1) \times k! + ((k+1)^2 + 1) \times (k+1)! = (k+1)((k+1)+1)! \quad \textcircled{2}$$

Proof: LHS of $\textcircled{2} = k(k+1)! + (k^2 + 2k + 1 + 1)(k+1)!$
 By assumption
 $= (k+1)! (k + k^2 + 2k + 2)$
 $= (k+1)! (k^2 + 3k + 2)$
 $= (k+1)! (k+2)(k+1) = (k+2)! (k+1) = \frac{(k+1)!}{(k+1)} (k+1)!$
 $\text{LHS} = \text{RHS}$

d) Cont.

$\therefore$ Statement true for $n = k+1$

(7)

Step 3 Hence, by mathematical induction statement true for all positive integers $n \geq 1$.

e)

$$\angle X = 35^\circ \text{ (given)}$$

$$\angle PQR = 25^\circ \text{ (given)}$$

$$\angle PRQ = 35^\circ = \angle X \text{ (Angles in same segment are equal)}$$

$$\therefore \angle RPQ = 120^\circ \text{ (Sum triangle RPQ = } 180^\circ \text{)}$$

$$\angle XPY = 90^\circ \text{ (}\angle \text{ in semi circle is right angle)}$$

$$\angle YPR = 90^\circ \text{ (adjacent, supplementary to } \angle XPY \text{)}$$

$$\angle YPR = \angle RPQ - \angle YPQ$$

$$= 120^\circ - 90^\circ$$

$$= \underline{\underline{30^\circ}}$$

$$\boxed{a} \quad a = 10 \text{ m}, \quad P = 10 \text{ sec}$$

$$\omega = \frac{2\pi}{P} = \frac{\pi}{5} \text{ rad/sec} \quad \left(\frac{1}{2}\right)$$

$$\textcircled{i} \quad x = 10 \cos(\omega t + \alpha)$$

$$t = 0, \quad x = 10 \text{ m} \Rightarrow \quad \left(\frac{1}{2}\right)$$

$$10 = 10 \cos \alpha \Rightarrow \alpha = 0$$

$$\Rightarrow x = 10 \cos \frac{\pi}{5} t$$

$$x = 6 \text{ m} \quad \left(\frac{1}{2}\right)$$

$$\Rightarrow 6 = 10 \cos \frac{\pi}{5} t$$

$$\Rightarrow t = \frac{5}{\pi} \cos^{-1} \frac{3}{5} \quad \left(\frac{1}{2}\right)$$

$$\textcircled{ii} \quad \dot{x} = -2\pi \sin \frac{\pi}{5} t \quad \left(\frac{1}{2}\right)$$

$$\text{The speed} = |\dot{x}|$$

$$= 2\pi \sin \frac{\pi}{5} \cdot \frac{5}{\pi} \cos^{-1} \frac{3}{5}$$

$$= 2\pi \times \frac{4}{5}$$

$$= \frac{8\pi}{5} \approx 5 \text{ m/s} \quad \left(\frac{1}{2}\right)$$

no penalty for $\pm 5 \text{ m/s}$

1

$$\boxed{b} \quad f(x) = \ln x + \sin x \quad \left(\frac{1}{2}\right)$$

$$f'(x) = \frac{1}{x} + \cos x$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \quad \left(\frac{1}{2}\right)$$

$$= 0.5 - \frac{\ln(0.5) + \sin(0.5)}{2 + \cos(0.5)}$$

$$= 0.5 - \frac{-0.2137216}{2.87758}$$

$$= 0.5742712$$

$$\approx 0.5743 \quad \left(\frac{1}{2}\right)$$

$$\approx 0.5743$$

2

OR

$$V^2 = n^2(a^2 - x^2)$$

$$= \frac{\pi^2}{25} (100 - 36)$$

$$= \frac{64\pi^2}{25}$$

$$V = \frac{8\pi}{5} \approx 5 \text{ m/s}$$

no penalty for $\pm 5 \text{ m/s}$

Qn (14): Continued

C (i) $\ddot{x} = 0$ (1)
 $\dot{x} = C$ (2)
 $\dot{x} = U \cos \alpha$ (3)

$$x = Ut \cos \alpha + C$$

$$t=0, x=0 \Rightarrow C=0$$

$$\Rightarrow x = Ut \cos \alpha \quad \text{--- (1)}$$

$$\ddot{y} = g \quad \text{--- (2)}$$

$$\dot{y} = -gt + C$$

$$t=0, \dot{y} = U \sin \alpha$$

$$\Rightarrow C = U \sin \alpha$$

$$\Rightarrow \dot{y} = U \sin \alpha - gt \quad \text{--- (3)}$$

$$y = Ut \sin \alpha - \frac{1}{2}gt^2 + C$$

$$t=0, y=0 \Rightarrow C=0 \quad \text{--- (4)}$$

$$\Rightarrow y = Ut \sin \alpha - \frac{1}{2}gt^2 \quad \text{--- (5)}$$

3

(9)
(ii) $t = \frac{x}{U \cos \alpha} \quad \text{--- (1)}$

Sub. into y,

$$y = (U \sin \alpha) \frac{x}{U \cos \alpha} - \frac{1}{2}g \left(\frac{x}{U \cos \alpha} \right)^2 \quad \text{--- (2)}$$

$$= x \tan \alpha - \frac{g x^2}{2 U^2 \cos^2 \alpha}$$

$$y = x \tan \alpha - \frac{g x^2}{2 U^2} \sec^2 \alpha \quad \text{--- (3)}$$

2

Qn(14): Continued

C-iii) $U = 7\sqrt{10}$ m/s

$x = 10$ m, $y = 15$ m

Substitute into the eqn of the path:

$$y = 10 \tan \alpha - \frac{9.8 \times 100}{2 \times 490} \sec^2 \alpha$$

$$\tan^2 \alpha - 10 \tan \alpha + 16 = 0$$

$$\tan \alpha = \frac{10 \pm \sqrt{100 - 4 \times 1 \times 16}}{2}$$

$$= 5 \pm 3$$

$$\tan \alpha = 8 \text{ or } \tan \beta = 2$$

C-iv) Let t_1 & t_2 be the projection time of the 2 pebbles.

$$t_1 = \frac{x}{U \cos \alpha}$$

$$t_2 = \frac{x}{U \cos \beta}$$

Time elapsed,

$$t_1 - t_2 = \frac{x}{U \cos \alpha} - \frac{x}{U \cos \beta}$$

$$= \frac{x}{U} (\sec \alpha - \sec \beta)$$

$$= \frac{10}{7\sqrt{10}} (\sqrt{65} - \sqrt{5})$$

$$= \frac{\sqrt{650} - \sqrt{50}}{7}$$